AI-ML Assignment – 7

Total Marks: 10

Topic

Customer Segmentation using K-Means Clustering and Principal Component Analysis (PCA)

📅 Submission Deadline

29 July 2026, 11:59 PM IST

📄 Assignment PDF

Download the assignment from the shared Google Drive folder:

Google Drive Folder

https://drive.google.com/drive/folders/14_OGcBqYcEHc0F8Zfkezeqgf7LFcoDgF?usp=drive_link

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📊 Dataset

Mall Customer Segmentation Dataset

Kaggle Link:

https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python

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Problem Statement

A shopping mall wants to divide its customers into different groups based on their annual income and spending behavior. The management plans to use these groups for targeted marketing campaigns.

Develop a **K-Means Clustering** model to segment customers and apply **Principal Component Analysis (PCA)** to visualize the clusters in two dimensions.

Assignment Tasks (Total: 10 Marks)

Task 1: Data Understanding (2 Marks)

1. Load the dataset using Pandas.
2. Display the first five records.
3. Identify:
    - Numerical features
    - Categorical features (if any)
4. Display the dataset information and summary statistics.

Task 2: Data Preprocessing (2 Marks)

Perform the following:

- Check for missing values.
- Remove unnecessary columns (e.g., CustomerID).
- Encode categorical variables if required.
- Standardize the numerical features using **StandardScaler**.

Task 3: Model Development (3 Marks)

Perform the following:

1. Use the **Elbow Method** to determine the optimal number of clusters.
2. Train a **K-Means Clustering** model using the selected value of **K**.
3. Assign cluster labels to each customer.
4. Apply **Principal Component Analysis (PCA)** to reduce the dataset to **2 principal components**.

Task 4: Visualization and Evaluation (2 Marks)

Generate the following:

- Elbow Curve
- Scatter plot showing customer clusters
- PCA visualization with cluster labels

Write **3–4 observations** explaining:

- The optimal number of clusters.
- How PCA helps visualize high-dimensional data.
- Characteristics of the identified customer groups.

Task 5: Conclusion (1 Mark)

Write a **100–150 word** conclusion covering:

- Key findings.
- Business applications of customer segmentation.
- One limitation of K-Means clustering.
- One advantage of PCA.

Submission Instructions

Create a **public GitHub repository** containing:

- `Assignment-7.ipynb` or `Assignment-7.py`
- `README.md` including:
  - Objective
  - Dataset Link
  - Libraries Used
  - Methodology
  - Results
  - Conclusion

Do not upload the dataset to GitHub unless its license explicitly allows redistribution. Instead, provide the Kaggle dataset link in your README.md.

Google Form Submission

Submit your assignment using the Google Form:

https://forms.gle/fFL2CFooc5Vb2MXq8

Fill in:

- Name
- Registration Number
- Application Number
- Batch Number
- **Assignment Number (Assignment -7)**
- Public GitHub Repository Link
- Email Address

Important Instructions

- **Deadline: 29 July 2026, 11:59 PM IST**
- Only the **first valid submission** will be accepted.
- Late submissions will be automatically rejected.
- Duplicate submissions will be automatically rejected.
- Future assignment submissions are not permitted.
- Ensure your GitHub repository remains **public** until evaluation is completed.
- You will receive an email confirming whether your submission has been **accepted** or **rejected**.